Problem 2

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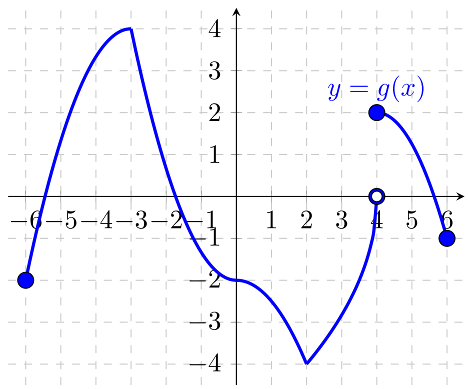
Problem 1 *The entire graph of a function g is given below.*



Based on the graph of g, answer the questions below.

(a) *List the x-coordinates of all critical points of g.*

Explanation. $x = -3$, $x = 0$, $x = 2$, and $x = 4$.

(b) *List the x-coordinates of all critical points of g where $g'(x) = 0$.*

Explanation. $x = 0$

(c) *List the x-coordinates of all critical points of g where $g'(x)$ is* ***undefined****.*

Explanation. $x = -3$, $x = 2$, and $x = 4$.

(d) *List the x-coordinates of all local maximums of g.*

Explanation. $x = -3$ and $x = 4$.

(e) *List the x-coordinates of all local minimums of g.*

Explanation. $x = 2$

(f) *List all intervals where g is both decreasing AND concave down.*

Explanation. $(0, 2)$ and $(4, 6)$.

(g) *List all intervals where g is both decreasing AND concave up.*

Explanation. $(-3, 0)$

(h) *List all intervals where g is both increasing AND concave down.*

Explanation. $(-6, -3)$

(i) *List all intervals where g is both increasing AND concave up.*

Explanation. $(2, 4)$

(j) *List the x-coordinates of all inflection points of g.*

Explanation. $x = -3$, $x = 0$, and $x = 2$.